WIRELESS NETWORKS

Wireless communication

Lecture 1

Fundamentals of wireless signals and networks

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Wireless communication

Devices exchange information using electromagnetic signals without a physical cable between them.



Encodes information

- Source and channel coding, then modulation
- Power amplifier drives the antenna

Carries the signal

- Attenuation, multipath and noise
- Interference from other transmitters

Recovers information

- Filtering, amplification, demodulation
- Error detection and correction

Example: a laptop exchanges data with a Wi-Fi access point.

Link quality is set by the signal-to-noise ratio: $SNR(dB) = 10 \cdot \log_{10}(P_signal / P_noise)$.

dB: decibel

Computing becomes personal and mobile



Earlier systems

Large, fixed equipment served specialist users.

- Mainframes (1960s–70s) were shared, room-sized computers
- Users reached them through wired terminals in one building

Portable devices

People access information while moving.

- Laptops, smartphones, wearables and IoT sensors
- Needs wireless links, long battery life and handover between cells

IoT: Internet of Things

Why wireless networks?

Access and mobility

Users connect while moving and collaborate across a space.

- Handover keeps a session alive as a user moves between access points
- Roaming extends access across buildings and operator networks

Deployment and growth

Add users and guest access with less endpoint cabling.

- Coverage grows by adding access points, not new cable runs
- Suits sites where cabling is hard: heritage buildings, events, factories



Installation cost (CAPEX) and operating cost (OPEX) can fall, depending on coverage and maintenance needs.

Milestones in wireless communication



1896

Marconi’s wireless telegraph patent

Spark-gap transmitter sends Morse code

1901

Transatlantic radio signal

Cornwall to Newfoundland, about 3,500 km

1960s

Communications satellites

Telstar (1962) relays live TV across the Atlantic

Later

Cellular, WLANs, ad hoc and sensors

1G in 1979 to 5G in 2019; IEEE 802.11 Wi-Fi from 1997

TV: television · WLAN: wireless local area network · IEEE: Institute of Electrical and Electronics Engineers · 1G–5G: first to fifth generation of cellular

Broadband wireless technology

Richer information

Higher data rates support images, audio and video.

- $\boldsymbol{Shannon\ limit}$: $\boldsymbol{C} = \boldsymbol{B} \cdot \log_2(\mathbf{1} + \boldsymbol{SNR})$
- Wider bandwidth **B** and higher SNR raise the maximum data rate

Faster deployment

Less endpoint cabling can make access easier to install.

- Fixed wireless access delivers home broadband over 4G/5G
- Microwave point-to-point links replace trenching for backhaul



Coverage, power and backhaul are still necessary.

Example: $B = 20\ MHz\ and\ SNR = 30\ dB\ (\times 1000)\ give\ C \approx 20\ MHz \times 10 \approx 200\ Mbps.$

Wireless technologies across the spectrum

Frequency increases across these radio bands

3–30 MHz	30–300 MHz	300 MHz–3 GHz	3–30 GHz	30–300 GHz
HF High Frequency	**VHF** Very High Frequency	**UHF** Ultra High Frequency	**SHF** Super High Frequency	**EHF** Extremely High Frequency
Shortwave **λ: 100–10 m** Ionospheric skywave gives global reach	**FM radio, television** **λ: 10–1 m** Line of sight with some diffraction	**Cellular, some sensor links** **λ: 1 m–10 cm** Good building penetration	**Wi-Fi, microwave links** **λ: 10–1 cm** High capacity; walls attenuate	**Millimetre-wave links** **λ: 10–1 mm** Wide bandwidth; rain and oxygen absorb

Infrared and optical links use still higher frequencies.

Examples may span several bands. This is not a frequency allocation chart.

MHz: megahertz (10^6 Hz) · GHz: gigahertz (10^9 Hz) · FM: frequency modulation

Evolution of cellular and Wi-Fi systems

Cellular



50 Mbps average*
LTE · OFDMA · up to 20 MHz per carrier

400 Mbps average*
NR · sub-6 GHz and mmWave · massive MIMO

Wi-Fi



300 Mbps*
802.11n · 2.4/5 GHz · MIMO

6.9 Gbps*
802.11ac · 5 GHz · 256-QAM

9.6 Gbps*
802.11ax · OFDMA · 1024-QAM

*Historical lecture examples mix average, configuration-dependent and theoretical rates. Actual throughput varies.

LTE: Long Term Evolution · OFDMA: Orthogonal Frequency-Division Multiple Access · NR: New Radio · MIMO: multiple-input multiple-output · QAM: quadrature amplitude modulation · mmWave: millimetre wave · Mbps / Gbps: megabits / gigabits per second

What improves wireless performance?

Hardware

Better batteries, circuits and processors.

- CMOS scaling puts a complete radio on one chip (RFIC)

Radio link

Antennas, modulation, coding, DSP and bandwidth.

- MIMO, higher-order QAM and LDPC/polar codes approach the Shannon limit

Network

More efficient sharing of radio resources.

- Smaller cells, frequency reuse, scheduling and OFDMA



CMOS: complementary metal-oxide-semiconductor · RFIC: radio-frequency integrated circuit · DSP: digital signal processing · LDPC: low-density parity-check

Wireless information access

Anytime
Anywhere
Any form

Ubiquitous computing makes information available throughout everyday environments.

- Smart homes, campuses and cities embed sensors and connectivity
- Users expect seamless handover between Wi-Fi and cellular

Different applications and coverage needs create a diversity of standards and products.

- **PAN:** Bluetooth, about 10 m
- **LAN:** Wi-Fi, tens of metres
- **WAN:** cellular, several kilometres
- Each trades data rate against range and power consumption

PAN: personal area network · LAN: local area network · WAN: wide area network

Challenges in wireless networks

Variable channel

Path loss, multipath fading and Doppler effects change reception.

- Free-space loss grows with distance^2 and frequency^2
- Doppler shift $f_d = v \cdot f / c$: 100 km/h at 2 GHz gives about 185 Hz

Shared radio medium

Interference, limited range and security threats need control.

- Users share one channel, so a MAC protocol must schedule access
- Anyone in range can receive the signal; encryption (WPA3) protects it



Data rates vary with conditions and may be lower than a comparable wired connection.

MAC: medium access control · WPA3: Wi-Fi Protected Access 3

Wireless around us



On the move

Cellular, satellite and broadcast services support mobile users.

- Macro cells cover kilometres; small cells add capacity in busy areas
- GNSS satellites (GPS, Galileo) orbit at about 20,000 km for positioning

Nearby devices

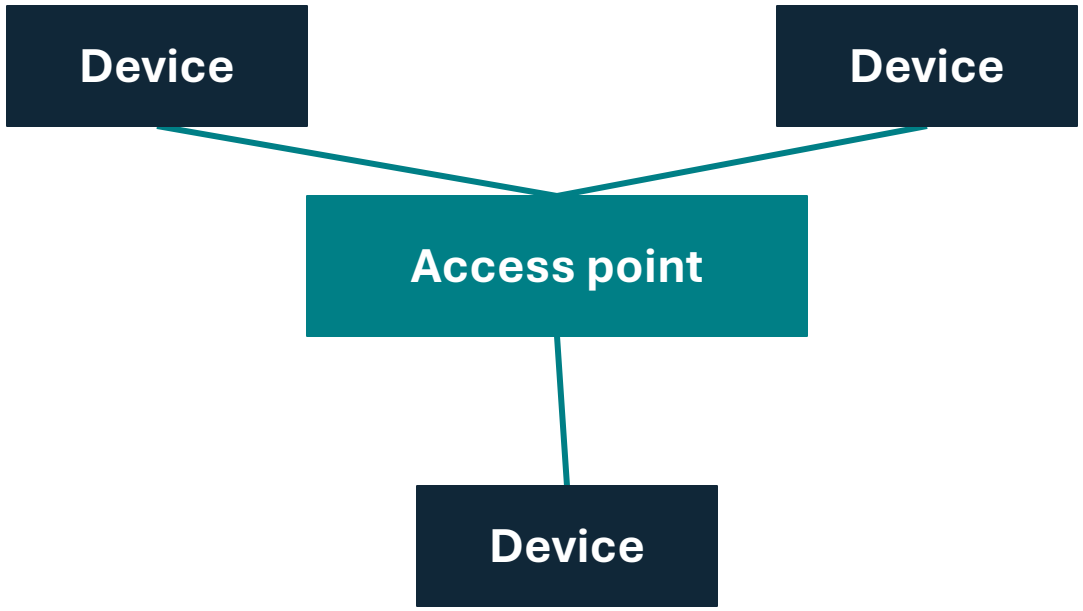
Wi-Fi and Bluetooth connect personal and local devices.

- Bluetooth Low Energy: 2.4 GHz, tens of metres, very low power
- NFC: 13.56 MHz over a few centimetres, used for payments

GNSS: Global Navigation Satellite System · GPS: Global Positioning System · NFC: near-field communication

Classification of wireless networks

Infrastructure-based



Wi-Fi or a cellular base station provides the access infrastructure.

- The network controls access, addressing and handover
- All traffic passes through the access point, even between nearby devices

Infrastructure-less



Direct: Wi-Fi Direct or Bluetooth

Devices link in one hop with no access point



Multihop: MANET, VANET or WSN

Nodes self-organise and relay for each other; no single point of failure

MANET: mobile ad hoc network · VANET: vehicular ad hoc network · WSN: wireless sensor network

Single-hop and multihop communication

Single hop



The destination is within direct radio reach.

- Low latency and simple routing
- Range set by transmit power and receiver sensitivity

Multiple hops



An intermediate device forwards the data.

- Extends coverage around obstacles
- Each hop adds delay and routing overhead

Example: a sensor forwards a measurement through another sensor.

Path loss grows as d^n ($n \approx 2–4$), so two hops of d/2 can need less total transmit power than one hop of d.

What actually travels between the devices?

Electromagnetic energy carries the information.

- An antenna turns current into a radiating wave
- No medium is needed: radio crosses air, vacuum and many materials
- Energy spreads over a sphere, so power density falls as $1/r^2$ in free space

Wireless signals are electromagnetic waves

Electric and magnetic fields

The fields vary in phase and are perpendicular to each other.

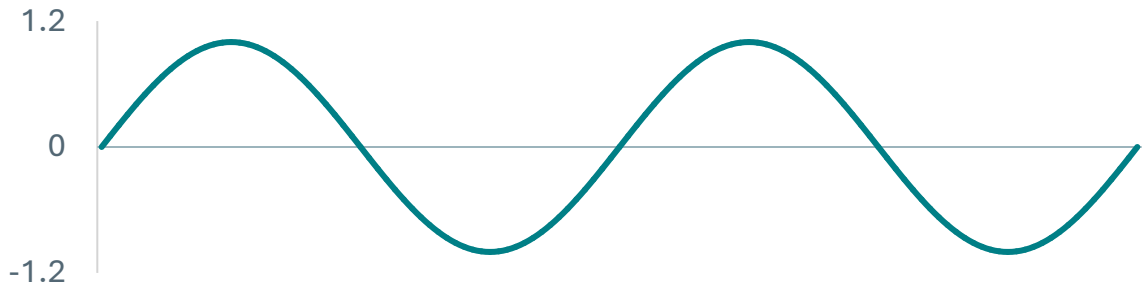
- In free space, $E = c \times B$ (E in V/m, B in tesla)
- The direction of E sets the polarization, so antenna orientation matters

Propagation

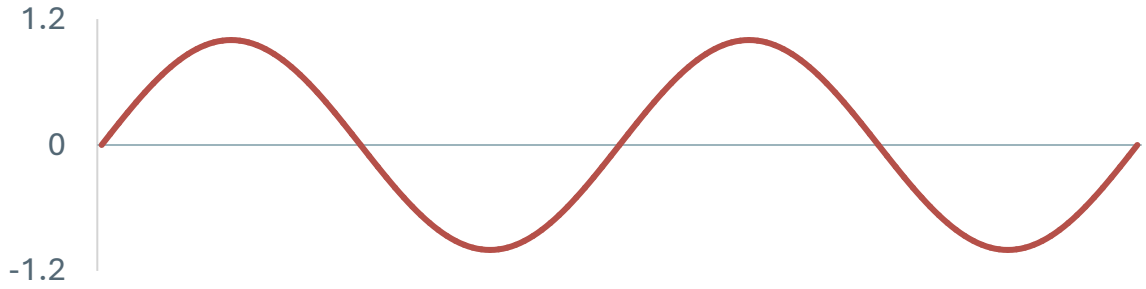
Both fields are perpendicular to the direction of travel.

- A transverse wave, so it needs no medium
- Speed in a vacuum: $c \approx 3 \times 10^8$ m/s

Electric field E

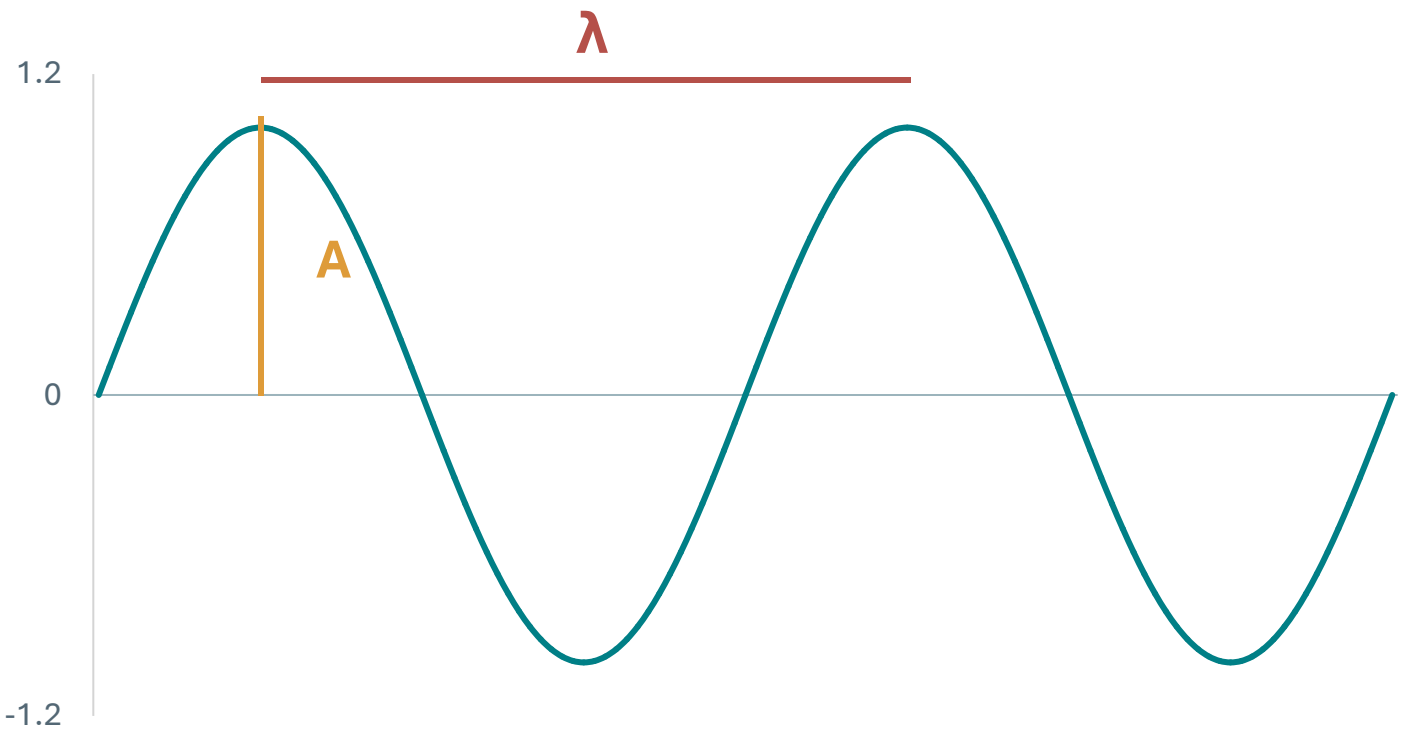


Magnetic field B



Normalized field projections along the same propagation direction.

Amplitude and wavelength



Spatial wave sketch: the horizontal direction represents distance.

Amplitude A

Maximum displacement from the zero level.

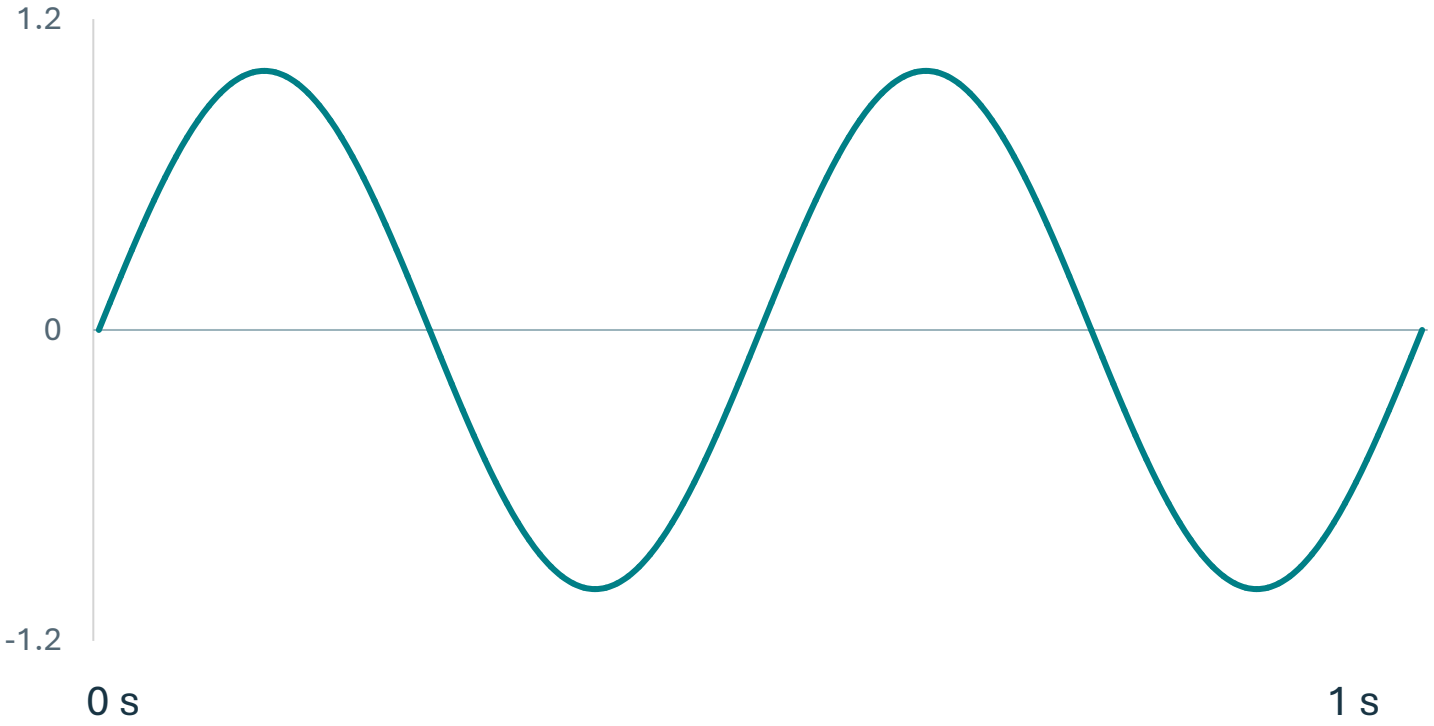
- Signal power is proportional to A^2
- Amplitude falls with distance from the transmitter

Wavelength λ

Distance between successive points in the same phase.

- Measured in metres
- Antenna size scales with λ: a half-wave dipole is λ/2 long

Frequency and period



$$f = 1 / T$$

Frequency f

Cycles per second, in hertz (Hz)

- 1 kHz = 10^3 Hz, 1 MHz = 10^6 Hz, 1 GHz = 10^9 Hz

Period T

Time for one cycle, in seconds

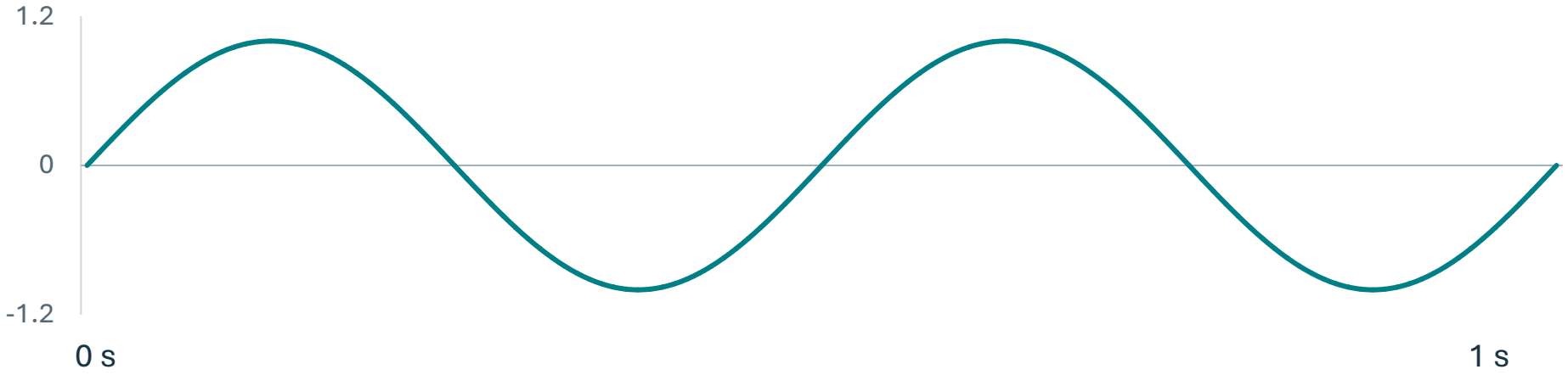
- At 2.4 GHz, $T = 1 / (2.4 \times 10^9) \approx 0.42$ ns

Two cycles in 1 second: f = 2 Hz and T = 0.5 s

Higher frequency means more cycles

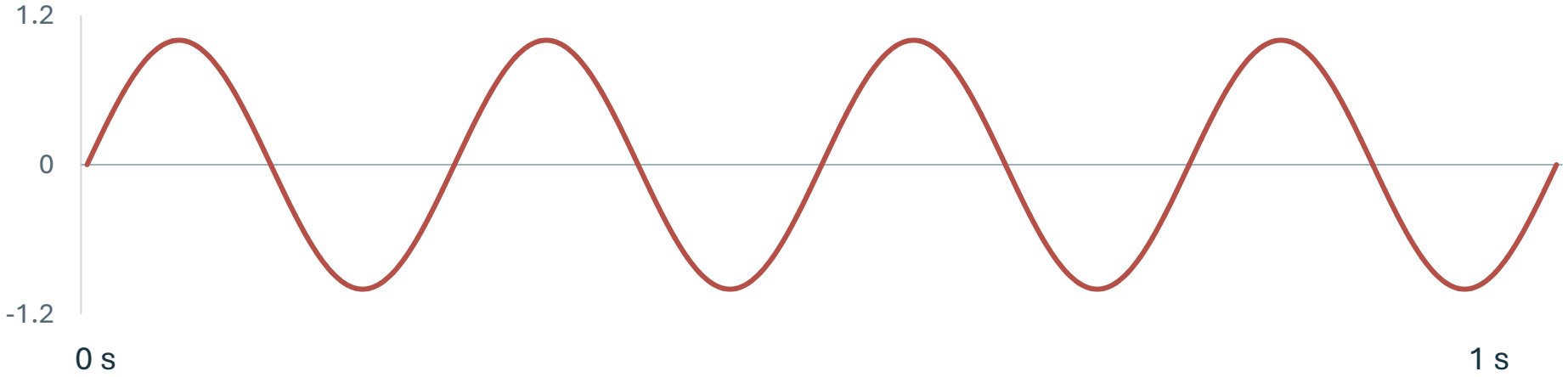
2 Hz

T = 0.5 s



4 Hz

T = 0.25 s



Same one-second interval and same amplitude. Only the frequency changes.

Since T = 1/f, doubling the frequency halves the period.

Frequency and wavelength

$$\lambda = c / f$$

$c \approx 3 \times 10^8$ m/s in air

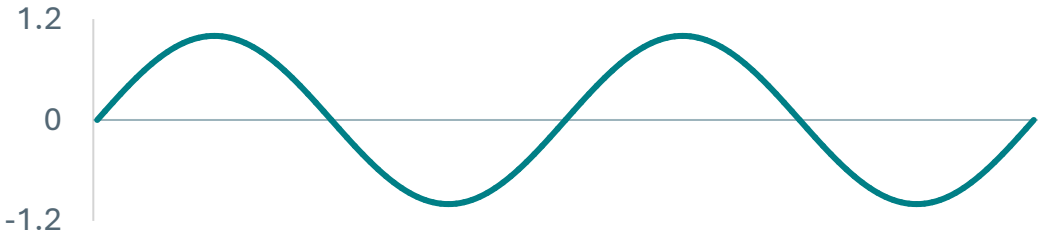
At 2.4 GHz

$\lambda = (3 \times 10^8) / (2.4 \times 10^9)$
$\lambda = 0.125$ m = 12.5 cm

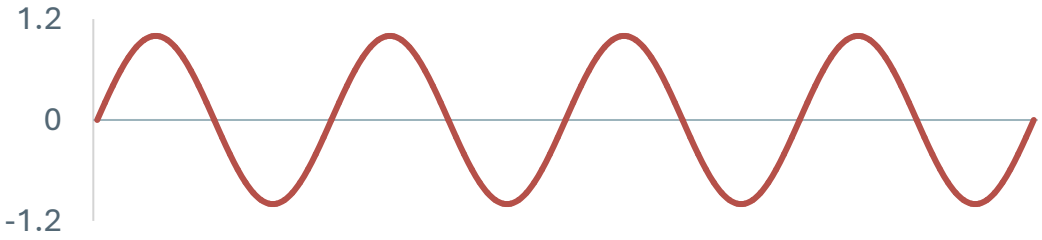
5 GHz Wi-Fi: λ = 6 cm, so a quarter-wave (λ/4) antenna is about 1.5 cm long

Same distance: 25 cm

2.4 GHz: λ = 12.5 cm



4.8 GHz: λ = 6.25 cm



Doubling frequency halves wavelength. **λ (m) = 300 / f (MHz)**

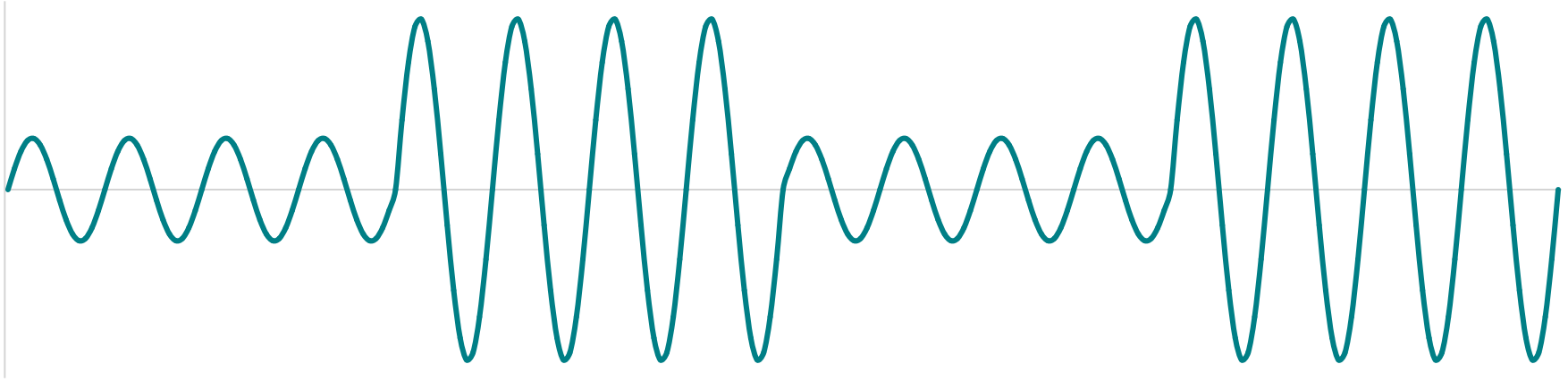
Modulation carries information

Modulation changes a carrier wave according to the information being sent.

The carrier is a high-frequency sinusoid: $s(t) = A \cdot \cos(2\pi f t + \phi)$, with amplitude A, frequency f and phase ϕ.

Information bits 0 1 0 1

Illustrative amplitude change



A signal can vary amplitude, frequency or phase to represent information.

ASK varies A, FSK varies f, PSK varies ϕ; QAM changes A and ϕ together to send several bits per symbol.

ASK: amplitude-shift keying · FSK: frequency-shift keying · PSK: phase-shift keying

The electromagnetic spectrum

Increasing frequency



Longer wavelength

Shorter wavelength

Visible light occupies only a small part of the spectrum.

Radio and microwaves (below about 300 GHz) carry most wireless networks; infrared and visible light serve remote controls and Li-Fi.

All electromagnetic waves travel at c in a vacuum. 1 Hz = 1 cycle/s

Li-Fi: light fidelity (data over visible light)

Frequency examples in everyday systems











AM radio

About 1 MHz

λ ≈ 300 m

FM radio

About 100 MHz

λ ≈ 3 m

TV and cellular

Multiple bands

About 0.5–3.8 GHz

Wi-Fi

2.4 and 5 GHz

Wi-Fi 6E adds 6 GHz

Satellite

Microwave bands

C, Ku, Ka: 4–40 GHz

Illustrative examples. Exact bands depend on the service, system and region.

AM: amplitude modulation

Quick understanding check

01 Network organization

A laptop connects through a Wi-Fi access point. Which category is this?

Answer: infrastructure-based. The access point provides the access infrastructure.

02 Frequency

A signal completes 10 cycles in one second. What are f and T?

Answer: f = 10 Hz and T = 1/f = 0.1 s.

03 Wavelength

If frequency doubles while wave speed stays constant, what happens to λ?

Answer: λ halves, because λ = c / f.

04 Modulation

Which changes in a carrier can represent information?

Answer: amplitude, frequency or phase (ASK, FSK, PSK; QAM combines A and ϕ).

05 Challenge A phone uses 2.4 GHz Wi-Fi. How long is a quarter-wave (λ/4) antenna? A bus at 100 km/h uses a 2 GHz link: estimate the Doppler shift.

Answer: $\lambda = 3 \times 10^8 / 2.4 \times 10^9 = 12.5$ cm, so $\lambda/4 \approx 3.1$ cm. Doppler: $v = 27.8$ m/s, $\lambda = 0.15$ m, so $f_d = v / \lambda \approx 185$ Hz.

Lecture 1 recap

Wireless networks

Enable mobility and flexible access, with channel and security challenges.

Limits: path loss, fading, Doppler shift and interference

Network organization

Infrastructure-based links, direct links and multihop relaying.

Single hop is simpler and faster; multihop extends range at the cost of delay

Wireless signals

Electromagnetic waves described by amplitude, frequency and wavelength.

$\lambda = c / f$ and $f = 1 / T$; 2.4 GHz gives $\lambda = 12.5$ cm

Information transfer

Modulation changes the carrier to represent information.

ASK, FSK, PSK and QAM; capacity $C = B \cdot \log_2(1 + SNR)$